



E-commerce Business Analysis using SQL & Python

End-to-End Data Preparation, Analytics & Business Insights

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Project Objective

This project demonstrates end-to-end analytical capabilities by transforming raw e-commerce data into strategic business intelligence.

Extract Insights

Analyze real-world e-commerce datasets to uncover actionable patterns in sales, customers, and products

Multi-Tool Approach

Leverage SQL for scalable analytics and Python for data preparation and rich visualizations

Drive Decisions

Support data-driven business strategy with clear metrics on performance, behavior, and growth

Questions Addressed

Our analysis answers critical business questions that shape strategic decisions and operational priorities.

Revenue Trends

How is revenue trending over time? Identifying growth patterns and seasonal fluctuations

Top Contributors

Which customers and sellers generate the most value? Pinpointing high-impact segments

Product Performance

What products drive volume versus revenue? Understanding the catalog's true performers

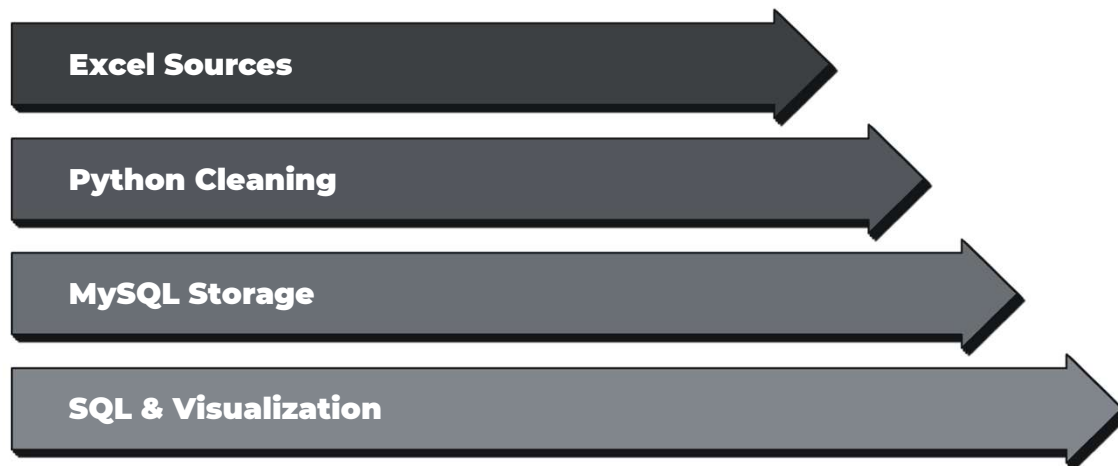
Pricing Impact

How does pricing affect customer demand? Analyzing price sensitivity and elasticity

Business Growth

What is the overall trajectory? Measuring expansion, retention, and long-term health

Tools & Workflow



This workflow mirrors real-world production analytics environments where different tools handle specialized responsibilities.

Python

Conver the dataset format, load the data into workbench, perform basic checks, and creates rich visualizations

MySQL

Scalable analytics through complex joins, aggregations, and window functions

GitHub

Share the all work along with python notebook, SQL script, and datasets

Dataset Overview

The provided dataset files spanning multiple dimensions and categories of the business, from customer demographics to payment behavior.

CUSTOMERS	Rows: 99441	Columns: 5
ORDERS	Rows: 99441	Columns: 8
ORDER_ITEMS	Rows: 112650	Columns: 7
PAYMENTS	Rows: 103886	Columns: 5
PRODUCTS	Rows: 32951	Columns: 9
SELLERS	Rows: 3095	Columns: 4
GEOLOCATION	Rows: 1000163	Columns: 5

Customers

Customer ID, Unique ID, Zip Code, City, and State information is present

Orders

Have the details of order status with time stamps. Order ID is there with respect to customer ID

Order Items

This has the product Id with the price and quantity, ordered with respect to the order ID

Payments

With respect to the order ID , it has the amount, method of payment and installments details

Products

This has details of products with the name, size, photos and description code

Sellers

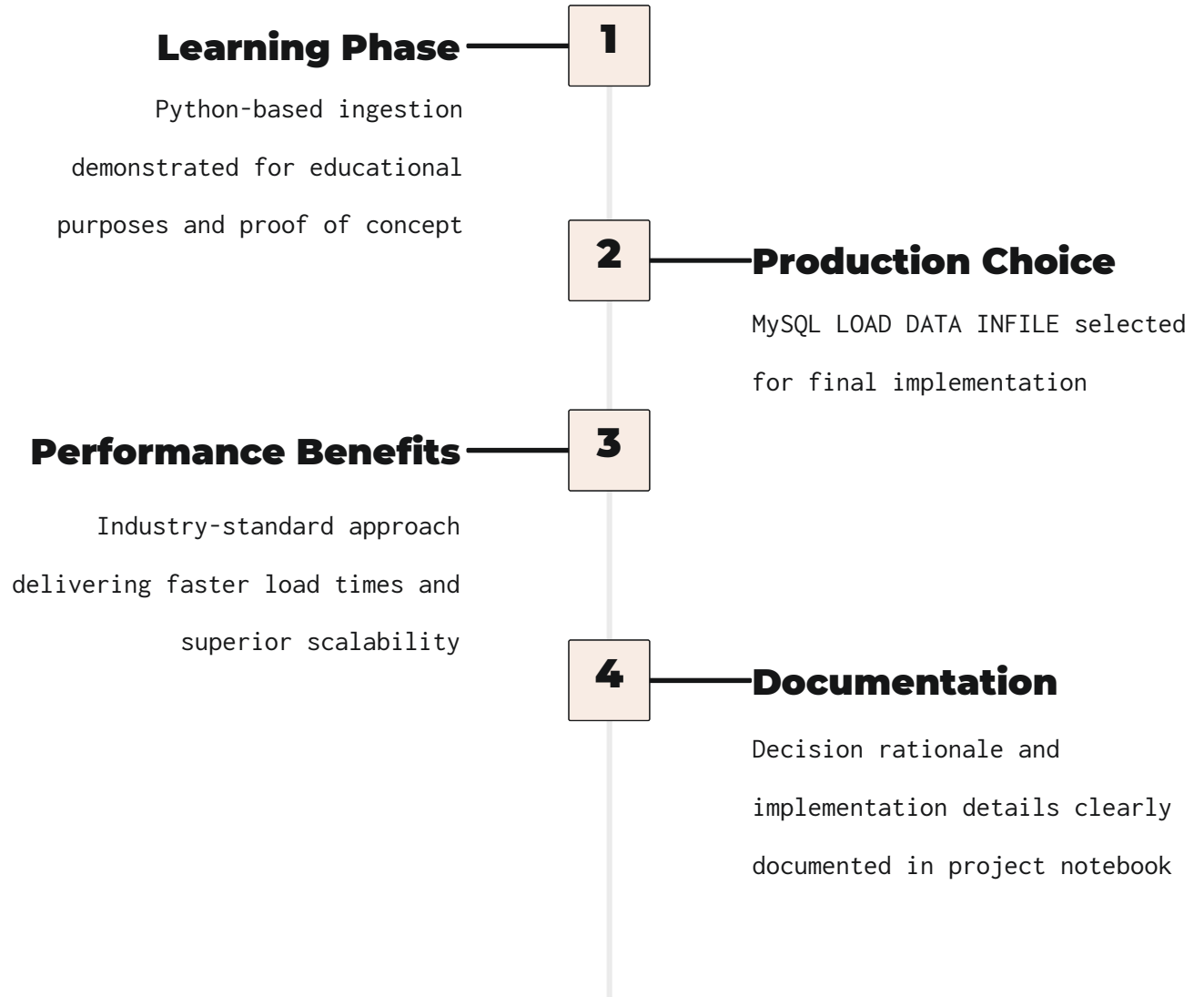
Information of seller location with the city, state and zip code

Geolocation

With respect to the Zip code it has the coordinates, city name and the state code.

Data Ingestion Strategy

Loading data efficiently is critical for production-scale analytics and sets the foundation for all downstream analysis.



SQL Analysis Scope

The analysis progresses through three tiers of increasing complexity, building from foundational metrics to sophisticated business intelligence.



Basic Analysis

Order counts, customer demographics, payment method distribution, and fundamental business metrics



Intermediate Analysis

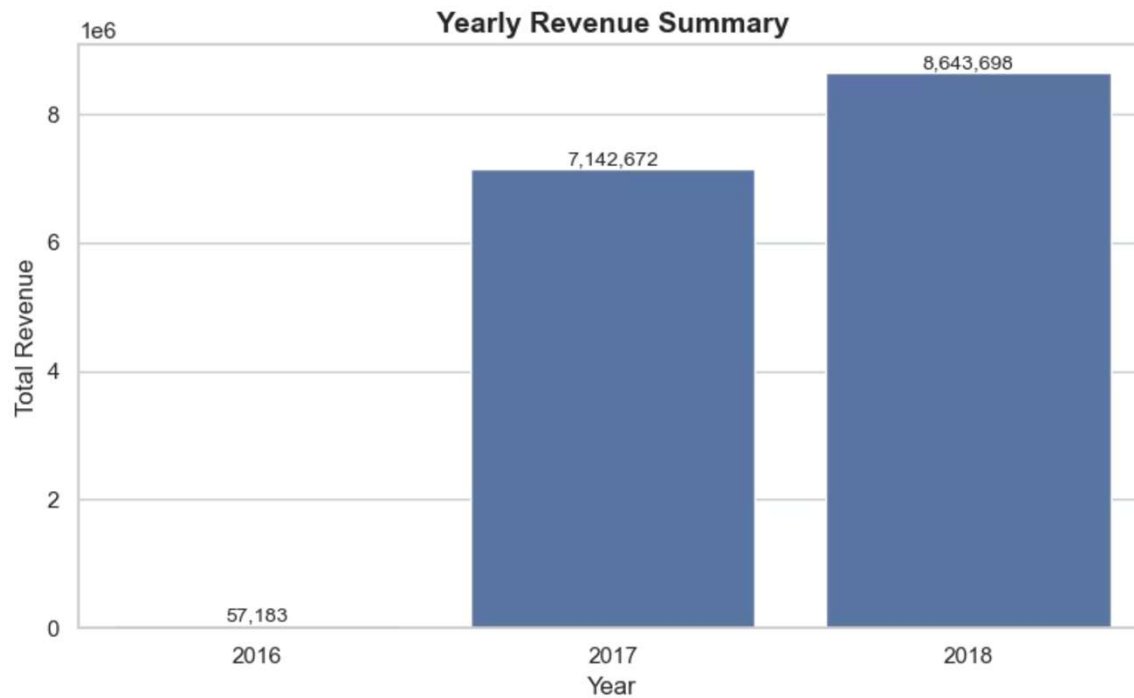
Revenue calculations, seller performance rankings, time-based trends, and comparative metrics



Advanced Analysis

Growth rate modeling, customer retention cohorts, lifetime value calculation, and top contributor identification

Revenue Performance



Strong Growth

- Year-wise revenue analysis shows that 2018 was the strongest performing year, generating a total revenue of 8,643,698.
- The revenue trend closely aligns with order growth, reinforcing volume-driven revenue expansion.

Customer Value Analysis

High Value order Customer

Our analysis reveals that 10 top customers who places the higher value order indicating the customer who spent high amount.

Actionable Opportunities:

- Targeting more similar customers
- Add the valuable and most selling products
- Help managing the product pricing

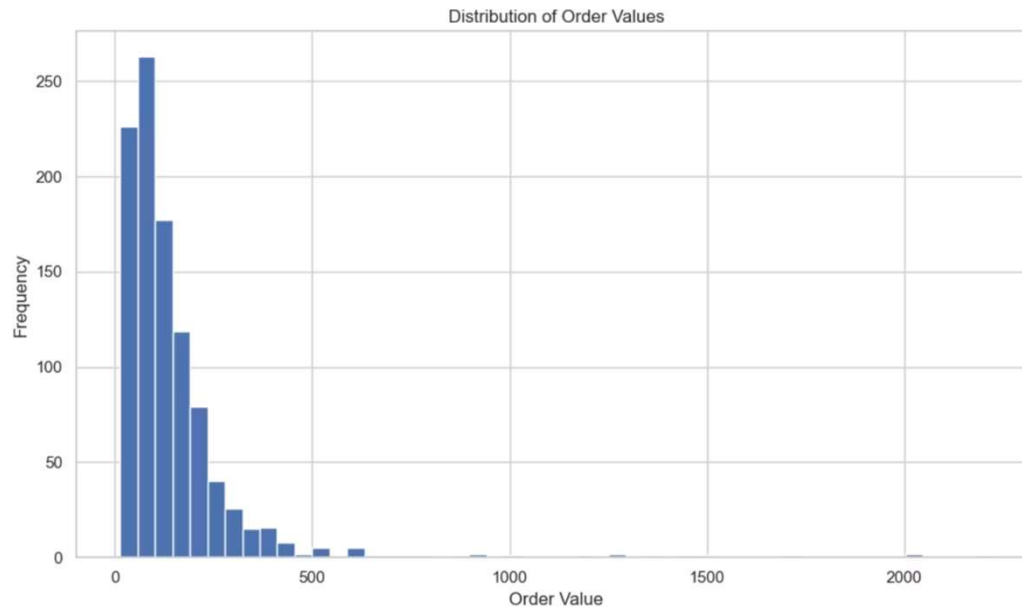


Customer Retention Analysis

- The dataset assigns a unique customer ID to every order, resulting in no identifiable repeat customers.
- As a result, the calculated customer retention rate is 0%.
- This highlights a data limitation, making retention analysis inconclusive rather than negative.



Order Value Distribution



Order Volume Trends

- Across 2016, 2017, and 2018, the platform received a total of 99,441 orders.
- Year-wise breakdown:
 - 2016: 329 orders
 - 2017: 45,101 orders
 - 2018: 54,011 orders
- Although **2018** recorded the highest number of orders, the increase from 2017 is **incremental** rather than **exponential**, indicating **steady and sustainable growth** rather than a sudden spike.

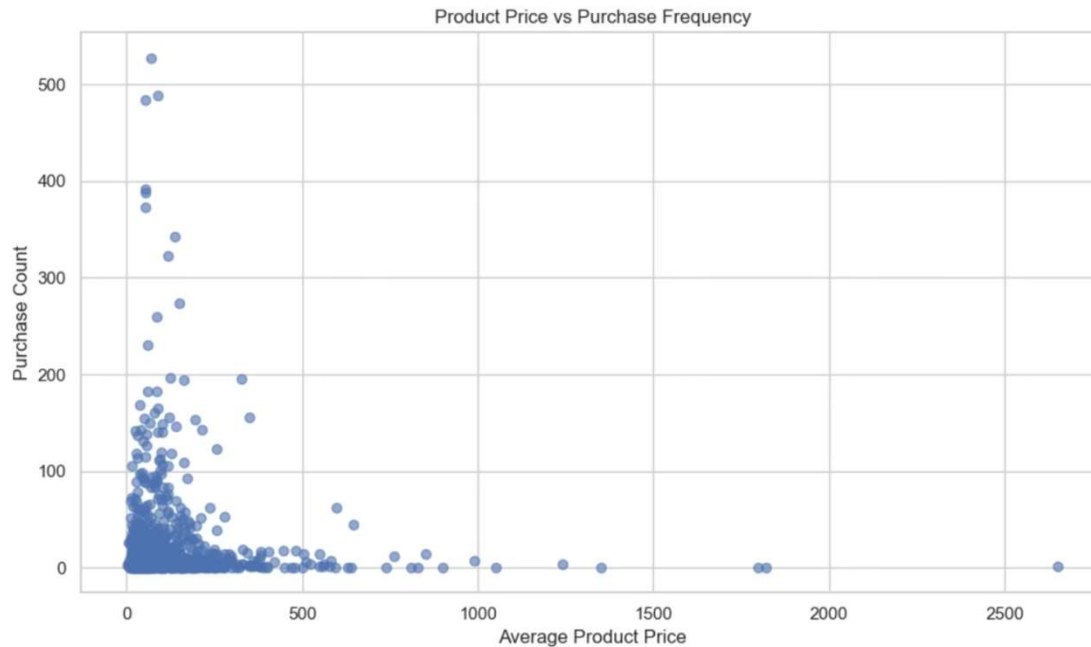
Volume Driver

Most orders are modest in value but high in frequency

Revenue Outliers

Few high-value transactions contribute significantly to total revenue

Pricing vs Demand Relationship



Classic Demand Curve

Our analysis confirms an inverse relationship between price and demand, consistent with fundamental economic principles.

Volume Products

Low-priced items drive transaction volume and customer acquisition

Premium Segment

High-priced products serve niche markets with lower velocity but higher margins

High-Value Product Insight

- Housewares emerged as the most expensive product category.
- Despite higher pricing, it recorded a substantial total quantity sold of 6,795 units, highlighting strong demand even at premium price points.



Seller Performance

- The marketplace consists of 3,095 active sellers.
- The top-performing seller (4869f7a5dfa277a7dca6462dcf3b52b2) completed:
 - ❑ 1,124 orders
 - ❑ Generated **226,987** in revenue, making it the highest-volume seller.
- The second-highest seller (53243585a1d6dc2643021fd1853d8905) delivered:
 - ❑ 348 orders
 - ❑ Generated **217,940** in revenue, indicating a high average order value and sale of premium-priced items.



Key Cross-Analysis Insights



Revenue Growth

Consistent year-over-year revenue growth indicates strong platform adoption, increasing order volume, and sustained business expansion



Customer Spending

Top 10 customers shows the order placed at higher value. But the retention rate of the customer is zero. The data has only one customer per unique id.



Pricing Strongly Affects Demand

Lower-priced products drive volume while premium items serve profitable niches, informing optimal pricing and inventory strategies.



SQL + Python Complement

Combining SQL's analytical power with Python's visualization capabilities creates a robust, scalable analytics framework for comprehensive insights.

Project Outcomes



End-to-End Analytics Pipeline

Built a comprehensive data workflow from ingestion through visualization, demonstrating practical application of modern analytics best practices.



Scalable SQL Analysis

Developed efficient, reusable SQL queries that handle complex business logic and can scale with growing data volumes.



Executive-Ready Visual Insights

Delivered executive-ready visual insights that translate technical analysis into clear, actionable business intelligence.



Well-Documented, Reproducible Project

Maintained comprehensive documentation and version control, ensuring the analysis is fully reproducible, auditable, and extensible.

THANK YOU